

PIL2-STARK: Query Phase

Prover	Verifier
<i>Setup: Transcript reconstruction</i>	Replay transcript from proof data to rederive all challenges $(\alpha, \gamma, v_c, \xi, v_1, v_2, \{\beta_k\}, \chi_{\text{grind}})$
<i>Constraint check</i>	Evaluate $C(\xi)$ from $\{e_{p,o}\}$ Reconstruct $Q(\xi) = \sum_j \xi^{jN} e_{Q_j,0}$ Check: $Q(\xi) = C(\xi)/Z_H(\xi)$
<i>Proof-of-work check</i>	Check: $\text{Poseidon2}(\chi_{\text{grind}} \parallel \eta)$ has b_{pow} leading zeros
<i>Final polynomial degree check</i>	$\hat{F}_K = \text{INTT}(F_K)$ Check: $\hat{F}_K[i] = 0$ for $i \geq D$
<i>Query derivation and verification</i>	
Both sides hold all commitments, challenges, and η	= Both sides hold all commitments, challenges, and η
$(q_1, \dots, q_{Q_{\text{queries}}})$	$\leftarrow$ Seed $\mathcal{T}'$ from $(\chi_{\text{grind}}, \eta)$
For each query q_i , and each commitment tree	$\rightarrow$ $(q_1, \dots, q_{Q_{\text{queries}}}) \leftarrow \mathcal{T}'.\text{squeeze_indices}(Q_{\text{queries}}, b_0)$
(stages $1, 2, Q$, constants, FRI layers $0, \dots, K-1$):	For each query q_i :
compute Merkle opening proof at the appropriate index	Check: all Merkle proofs verify against committed roots Let $x_{q_i} = g \cdot \omega_{\text{ext}}^{q_i}$ Check: $F(x_{q_i})$ from batching formula matches FRI layer 0 at q_i For each FRI round $k = 1, \dots, K$: Interpolate coset siblings Apply coset correction Evaluate at $\hat{\beta}_k$ Check: result matches layer k Accept iff all checks pass